IEOR 221 - HW 10 - Jie Wen

1

Using the single-factor model formulas:

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + e_i^2,$$

where e_i is the firm-specific (idiosyncratic) standard deviation and σ_M is the market standard deviation.
Here, $\sigma_M = 22\% = 0.22$.

a. Standard deviations of A and B

Stock A

With $\beta_A = 0.8$ and firm-specific sd $e_A = 30\% = 0.30$:

$$\sigma_A^2 = (0.8)^2(0.22)^2 + (0.30)^2 \approx 0.12098$$

$$\sigma_A \approx \sqrt{0.12098} \approx 0.348 \quad \Rightarrow \quad \boxed{\sigma_A \approx 34.8\%}$$

Stock B

With $\beta_B = 1.2$ and firm-specific sd $e_B = 40\% = 0.40$:

$$\sigma_B^2 = (1.2)^2(0.22)^2 + (0.40)^2 \approx 0.22970$$

$$\sigma_B \approx \sqrt{0.22970} \approx 0.479 \quad \Rightarrow \quad \boxed{\sigma_B \approx 47.9\%}$$

b. Portfolio with A (30%), B (45%), Risk-free (25%)

Weights: $w_A = 0.30,\ w_B = 0.45,\ w_f = 0.25$.

$$E[R_P] = \sum w_i \mu_i = 0.30(0.13) + 0.45(0.18) + 0.25(0.08) = \boxed{14\%}$$

The risk-free asset has $\beta_f = 0$.

$$\beta_P = w_A\beta_A + w_B\beta_B + w_f\beta_f = 0.30(0.8) + 0.45(1.2) + 0.25(0) = \boxed{0.78}$$

From the single-factor model, the total variance is composed of systematic and idiosyncratic components:

$$\sigma_P^2 = \underbrace{\sum_{i,j} w_i w_j \beta_i \beta_j \sigma_M^2}_{\text{systematic}} + \underbrace{\sum_i w_i^2 e_i^2}_{\text{idiosyncratic}}$$

The nonsystematic (idiosyncratic) variance is:

$$\sigma_{\text{ns},P}^2 = w_A^2 e_A^2 + w_B^2 e_B^2$$

$$w_A^2 e_A^2 = (0.30)^2(0.30)^2 = 0.09 \cdot 0.09 = 0.0081$$

$$w_B^2 e_B^2 = (0.45)^2(0.40)^2 = 0.2025 \cdot 0.16 = 0.0324$$

$$\sigma_{\text{ns},P}^2 = 0.0081 + 0.0324 = 0.0405$$

$$\sigma_{\text{ns},P} = \sqrt{0.0405} \approx 0.201 \quad \Rightarrow \quad \boxed{\sigma_{\text{ns},P} \approx 20.1\%}$$

The systematic (market-driven) component is:

$$\sigma_{\text{sys},P} = \beta_P \sigma_M = 0.78 \cdot 0.22 = 0.1716 \quad (\approx 17.2\%)$$

The total variance is the sum of systematic variance and nonsystematic variance:

$$\sigma_P^2 = \beta_P^2 \sigma_M^2 + \sigma_{\text{ns},P}^2 = (0.78)^2(0.22)^2 + 0.0405$$

$$(0.78)^2 = 0.6084$$

$$(0.22)^2 = 0.0484$$

$$0.6084 \cdot 0.0484 \approx 0.02945$$

$$\sigma_P^2 \approx 0.02945 + 0.0405 = 0.06995$$

$$\sigma_P \approx \sqrt{0.06995} \approx 0.264 \quad \Rightarrow \quad \boxed{\sigma_P \approx 26.4\%}$$

2

Using the CAPM formula:

$$\mu_i = r + \beta_i(\mu_M - r),$$

where $r = 6\% = 0.06$ and $\mu_M = 16\% = 0.16$.

a. Expected end-of-year stock price

Given:

- $S_0 = 50$
- Dividend at year end: $D_1 = 6$
- $\beta = 1.2$

$$\mu_i = 0.06 + 1.2(0.16 - 0.06) = 0.18.$$

Total expected return over the year:

$$\mu_i = \frac{D_1 + P_1 - S_0}{S_0}$$

Plug in numbers:

$$0.18 = \frac{6 + P_1 - 50}{50} \Rightarrow P_1 = \boxed{53}$$

b. Mispricing a firm due to underestimating beta

Perpetual cash flow: $C = 100{,}000$ per year.

- Assumed $\beta = 0.5$
- True $\beta = 1.0$

The discount rate (required return) comes from CAPM:

$$k(\beta) = r + \beta(\mu_M - r).$$

$$k_{0.5} = 0.06 + 0.5(0.16 - 0.06) = 0.11.$$

Value he *offers* (perpetuity):

$$V_{0.5} = \frac{C}{k_{0.5}} = \frac{100{,}000}{0.11} \approx 909{,}090.91.$$

$$k_{1.0} = 0.06 + 1.0(0.16 - 0.06) = 0.16.$$

True value:

$$V_{1.0} = \frac{100{,}000}{0.16} = 625{,}000.$$

$$\text{Overpayment} = V_{0.5} - V_{1.0} \approx 909{,}090.91 - 625{,}000 = \boxed{284{,}090.91}.$$

c. Beta of a stock with expected return 4%

Given:

- Expected return: $\mu_i = 4\% = 0.04$
- Still $r = 0.06,\ \mu_M = 0.16$

From CAPM:

$$\mu_i = r + \beta_i(\mu_M - r)$$

$$0.04 = 0.06 + \beta_i(0.16 - 0.06) \Rightarrow \beta_i = \boxed{-0.2}.$$

3

a. Investment decision

The CAPM expected return is calculated as:

$$\mu_{\text{CAPM}} = r + \beta(\mu_M - r)$$

Plugging in the numbers:

$$\mu_{\text{CAPM}} = 5\% + 0.8(15\% - 5\%) = 5\% + 0.8 \times 10\% = 5\% + 8\% = 13\%.$$

The fund's **actual expected return is 14%**, which is **1% higher** than the 13% required by CAPM for its risk level ($\beta = 0.8$).

- This means the fund has **positive alpha**:

$$\alpha = 14\% - 13\% = 1\% > 0$$

- Therefore, **yes**, under CAPM one should invest in this fund, because it offers more expected return than required for its systematic risk.

b. Passive portfolio with the same beta

Consider a portfolio combining the **market index ($\beta = 1$)** and the **risk-free asset ($\beta = 0$)**.

Let y be the fraction of wealth invested in the market index, and $1 - y$ in the risk-free asset.

The portfolio beta is:

$$\beta_P = y \cdot 1 + (1 - y) \cdot 0 = y$$

To match the fund's beta:

$$\beta_P = 0.8 \Rightarrow y = 0.8$$

So the passive portfolio is:

- **80% in the market-index portfolio**
- **20% in the risk-free (money market) account**

Its expected return would be:

$$E[R_P] = 5\% + 0.8(15\% - 5\%) = 13\%$$

This passive portfolio has **the same beta (0.8)** but a **lower expected return (13%)** than the mutual fund (14%), reinforcing that the fund looks attractive under CAPM.